COH-PRESERVING EXACT FUNCTORS COME FROM COHERENT KERNELS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Coh functor classification. functors-lemma-functor-coherent-over-field (cross-chapter, named). Produces the coherent kernel K flat over X with finite support over Y.
- Bounded coherence. lemma-fourier-mukai-Coh L26457 (carded this batch). F^prime preserves bounded coherence.
- Sibling transfer. lemma-sibling-fully-faithful L26481 (carded this batch). F and F^prime agree on Coh, hence are siblings.

Source reference. Original reference 0FZN: focused-seeds/geometry/S9-eqv-exact-functor-preserving-Coh/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYU: sources/stacks-equiv/pinned-original-source.tex, source lines 1355–1355. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZU: sources/stacks-equiv/pinned-original-source.tex, source lines 2208–2208. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X, Y be finite type schemes over k with X separated. Let $F : D^b_{Coh}(\mathcal{O}_X) \to D^b_{Coh}(\mathcal{O}_Y)$ be a k-linear exact functor sending $Coh(\mathcal{O}_X) \subset D^b_{Coh}(\mathcal{O}_X)$ into $Coh(\mathcal{O}_Y) \subset D^b_{Coh}(\mathcal{O}_Y)$. Then there exists a Fourier-Mukai functor $F' : D^b_{Coh}(\mathcal{O}_X) \to D^b_{Coh}(\mathcal{O}_Y)$ whose kernel is a coherent $\mathcal{O}_{X\times Y}$-module $\mathcal{K}$ flat over X and with support finite over Y which is a sibling of F.*

Proof. Denote $H : Coh(\mathcal{O}_X) \to Coh(\mathcal{O}_Y)$ the restriction of F. Since F is an exact functor of triangulated categories, we see that H is an exact functor of abelian categories. Of course H is k-linear as F is. By Functors and Morphisms, Lemma 0FZN we obtain a coherent $\mathcal{O}_{X\times Y}$-module $\mathcal{K}$ which is flat over X and has support finite over Y. Let F' be the Fourier-Mukai functor defined using $\mathcal{K}$ so that F' restricts to H on $Coh(\mathcal{O}_X)$. The functor F' sends $D^b_{Coh}(\mathcal{O}_X)$ into $D^b_{Coh}(\mathcal{O}_Y)$ by

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Lemma 0FYU. Observe that F and F' satisfy the first and second condition of Lemma 0FZU and hence are siblings. $\square$